

8. Factor: fertilizer type

Levels: 3

9. Factor: daily dose of Vitamin C

Levels: 3

14. data 有上下界 (0~1), normal $\rightarrow (-\infty, \infty)$ 且 data 是離散比例
 $\Rightarrow$ 不願意直接假設常態

8. Sample = 11, $\bar{X} = 76.364$

(a)

$$SSTR = \sum (\bar{X}_i - \bar{X})^2 = 641.879$$

$$SSE = A: 138$$

$$B: 232.667$$

$$C: SSE = 448.667$$

$$SST = SSTR + SSE = 641.879 + 448.667$$

$$= 1140.546$$

treatment: 2

error: 8

total: 10

$$MSTR = \frac{SSTR}{df_r} = \frac{641.879}{2} = 320.937$$

$$MSE = \frac{SSE}{df_e} = \frac{448.667}{8} = 62.333$$

$$F = \frac{MSTR}{MSE} = \frac{320.937}{62.333} = 5.149$$

	SS	df	MS	F
Treatment	641.879	2	320.939	5.149
Error	448.667	8	62.333	
Total	1140.546	10		

$$(b) H_0: \mu_A = \mu_B = \mu_C$$

H_a : 至少有一組平均數不同

$$F_{0.05, 2, 8} \approx 4.47 < F = 5.149$$

$\therefore$ reject H_0 .

$$10. (a) F = 5.704, p = 0.0251$$

$p = 0.0251 < 0.05$, 所以拒絕 H_0 .

(b) 99% CI

$$= (60.5 - 54.667) \pm 3.250 \sqrt{14.941 \left(\frac{1}{4} + \frac{1}{3} \right)}$$

$$= 5.833 \pm 9.59$$

$$= (-3.76, 15.42)$$

(c) Program A 99% CI

$$\bar{X}_A \pm t_{0.005, 3} \cdot \frac{S_A}{\sqrt{n_A}}$$

$$= 60.5 \pm 5.841 \cdot \frac{\sqrt{9.667}}{2}$$

$$= (51.42, 69.58)$$

(d) 常態假設是合理的

14. (a) ANOVA Table

	SS	df	MS	F
Treatment	0.922	3	0.307	1.48
Error	1.811	33	0.055	
Total	2.733	36		

$$(b) H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$$

$$H_a: \text{至少一组不同}$$

$$T = 1.490 < 2.892 < 4.337$$

$$\alpha = 0.05, \text{ Non-reject}$$

$$\alpha = 0.01, \text{ Non-reject}$$

$$(c) \overline{MSE} = \overline{X}_1 - \overline{X}_4 = 0.20638$$

$$SE = \sqrt{MSE(\frac{1}{n_1} + \frac{1}{n_4})} = \sqrt{0.20638(\frac{1}{19} + \frac{1}{11})} = 0.183$$

$$CI = (-0.6232, 0.1215) \text{ 不包含 } 0 \rightarrow \text{Non-reject}$$

$$(d) CI = 0.053 \pm 2.0345 \sqrt{0.20638(\frac{1}{8} + \frac{1}{4})}$$

$$= (-0.511, 0.621) \text{ 不包含 } 0 \rightarrow \text{non-reject}$$